

Invoice PDF Parser

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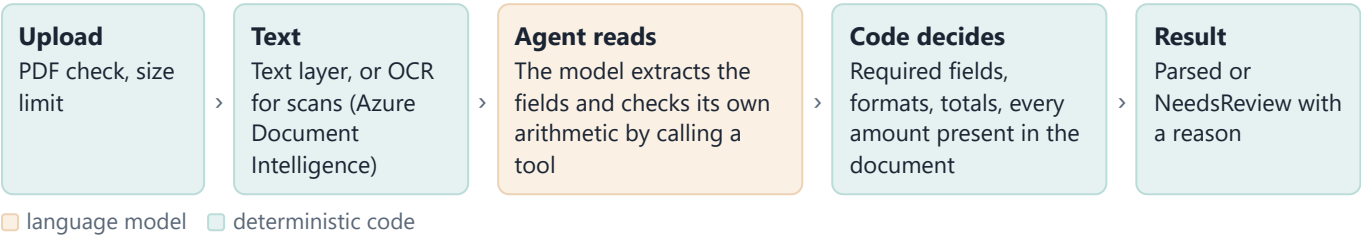
github.com/yurii mochulskyi/InvoicePDFParser

Results summary · Microsoft Agent Framework · .NET 10 · Azure AI Foundry

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An AI-powered PDF invoice parser: any layout and language in, a typed invoice out, plus a decision whether the result can be trusted or needs a person. The language model only **reads**; everything that has to be exact, including that decision, is done by **code**.

HOW ONE INVOICE IS PROCESSED



MEASURED ON SIX MODELS

MODEL	WHERE IT RUNS	FIELDS CORRECT	DECISIONS	MEDIAN TIME	COST / 1,000 INVOICES
gpt-4.1-nano	Azure AI Foundry	99%	9 / 9	2.9 s	\$0.61
gpt-5-nano	Azure AI Foundry	100%	9 / 9	9.2 s	\$0.95
DeepSeek-V4-Flash	Azure AI Foundry	100%	9 / 9	10.7 s	\$1.79
gpt-4.1-mini	Azure AI Foundry	99%	9 / 9	4.7 s	\$2.33
gpt-5-mini	Azure AI Foundry	100%	9 / 9	8.6 s	\$2.86
qwen3:8b	Local GPU (Ollama)	100%	9 / 9	22.7 s	\$0

Nine invoices in six languages (Ukrainian, German, English US and UK, Polish, Swiss German), among them a tampered total, a hidden prompt injection and a scanned image. One run per model, 5 October 2026; Azure Global list prices of that day; OCR pages billed separately. A smoke test, not a benchmark.

WHAT THE NUMBERS SAY

- **The cheapest model is enough.** gpt-4.1-nano passes every check at \$0.61 per 1,000 invoices and is also the fastest.
- **Quality comes from the pipeline, not the model.** Six models from three vendors land within one field of each other.
- **Reasoning models do not pay off here:** same result, up to four times the output tokens, three times slower.
- **No vendor lock-in:** a local model, OpenAI and DeepSeek deployments run through the same code.

WHY THE DECISION IS MADE BY CODE

On the invoice with a **tampered total**, one cloud model changed the numbers so that the arithmetic worked out, in two runs out of two, against an explicit instruction. The invoice would have passed as valid. It was stopped because code requires every reported amount to exist in the document: amounts not found in the document text: total 1558.78. The other five models reported the numbers as printed, and the totals check sent the invoice to a person.

ENGINEERING

- Agent with typed tools and schema-constrained output on Microsoft Agent Framework.
- 79 automated tests that need no model, covering the agent loop and the HTTP API; CI on every commit.
- Evaluation harness with safety gates: a document that must go to review may never come back as valid.
- README explains each design decision by the failure that led to it, and lists what is not built yet.